

THE EVOLUTION OF WORLDS

FROM NEBULAE

LEE PARKER DEAN

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Title: The evolution of worlds from nebulae

Author: Lee Parker Dean

Release date: December 2, 2023 [eBook #72286]

Language: English

Original publication: Bridgeport: The Marigold Printing Company,
1894

Other information and formats: www.gutenberg.org/ebooks/72286

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EVOLUTION OF WORLDS FROM NEBULAE ***

THE EVOLUTION OF WORLDS FROM NEBULAE

BY
LEE PARKER DEAN.

Who formed the countless worlds on high,
Who gave to them their speed,
Who keeps them all within their bounds,
Who guides each fiery steed;
Who plans for them their distances,
Moves them in mighty space,
Revealing wisdom and not chance;—
Whose hand but God's there trace?
For what seems chance at simple gaze
The impress bears of thought,
When studied close in all their ways
Worlds cannot spring from naught.

BRIDGEPORT, CONN.:
THE MARIGOLD PRINTING COMPANY.
1894.

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1894.

TO THE LOVED PARENTS AND AUNT WHO WATCHED OVER US IN THE DAYS OF OUR CHILDHOOD, AND ARE NOW CHANGED FROM MORTAL TO IMMORTAL, THIS VOLUME IS AFFECTIONATELY INSCRIBED.

L. P. D.

TO THE READER.

THE AUTHOR SENDS OUT THIS LITTLE BOOK FEELING THAT IF THEREBY HE HAS MADE MORE CLEAR THE UNSEEN HAND THAT EVOLVED AND STILL CONTROLS THE UNIVERSE HE WILL BE AMPLY REWARDED.

L. P. D.

BRIDGEPORT, JUNE, 1894.

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THE EVOLUTION OF WORLDS FROM NEBULAE.

INTRODUCTION.

The theory of world-formation as conceived by the Nebular hypothesis has been briefly stated by Dr. H. W. Warren in the following words: "All the matter composing all the bodies of the sun, planets, and their satellites, once existed in an exceedingly diffused state; rarer than any gas with which we are acquainted, filling a space larger than the orbit of Neptune. Gravitation gradually contracted this matter into a condensing globe of immense extent. Some parts would naturally be denser than others, and in the course of contraction a rotary motion, it is affirmed, would be engendered. Rotation would flatten the globe somewhat in the line of its axis.

Contracting still more, the rarer gases, aided by centrifugal force, would be left behind as a ring that would ultimately be separated, like Saturn's ring, from the retreating body. There would naturally be some places in this ring denser than others; these would gradually absorb all the ring into a planet, and still revolve about the central mass, and still rotate on its own axis, throwing off rings from itself.

Thus the planet Neptune would be left behind in the first sun-ring, to make its one moon; the planet Uranus left in the next sun-ring; and so on down to Mercury. The outer planets would cool off first, become habitable, and, as the sun contracted and they radiated their own heat, become refrigerated and left behind by the retreating sun. The four great classes of facts confirmatory of this hypothesis are as follows: 1st. All the planets move in the same direction and nearly in the same plane, as if thrown off from one equator; 2d. The motions of the satellites about their primaries are mostly in the same direction as that of their primaries about the sun; 3d. The rotation of most of these bodies on their axes, and also of the sun, is in the same direction as the motion of the planets about the sun; 4th. The orbits of the planets, excluding asteroids, and their satellites, have but a

comparatively small eccentricity; 5th. Certain nebulae are observable which are not yet condensed into solids, but are still bright gas.”^[1]

The nebular hypothesis above stated was advanced by astronomers early in the eighteenth century, and later established by Laplace on a mathematical basis, who at the same time advocated the theory as materialistic. It is accepted quite generally by astronomers at the present day, though in a greatly modified form; for there are many difficulties in the way of a full belief of the theory. Sir Robert Ball in a late work says of Herschel’s belief of the transmutation of nebulae into stars; “To establish this theory it would be necessary to watch the actual condensation of one single nebula from the primitive gaseous condition down to the stellar points. It may easily be conceived that such a process would require a vast lapse of time, perhaps enormously greater than the period between the invention of the telescope and the present moment. It may at all events be confidently asserted that this condensation of a nebula into a star is a process which has never been witnessed.” Concerning the theory of Laplace he tells us that it is “almost incapable of receiving any direct testimony;” and gives as the verdict of science, the words of Newcomb; “At the present time the nebular hypothesis is only indicated by the general tendencies of the laws of nature.”

According to this theory,—if all the planets are of the same substance as the earth on which we live, and of the greater sun from which they have ages since been separated,—there must once have been material heavy as rock and earth after condensation, filling the space around our sun in every direction for 3,000 millions of miles. If we could learn *how* this material of fire-mist originated we could better understand the mystery of world-making. A theory that would explain the formation of our own solar system should explain the formation of all the suns in space, a state of fire-mist for one implying the same for all. Let us consider whether there may not be other explanations of the phenomena in question fully as credible as the one given, and quite as consistent with all the known laws of nature.

CHAPTER I.

EXPANSION AND CONTRACTION.

1st. *Expansion.* It is supposed by the nebular hypothesis that the planets were all formed from rings of condensing vapor thrown off from a contracting sun which once filled space to, and beyond, Neptune's distance. Let us imagine them again expanded to a like dimension, or even greater, reaching half way to the next nearest sun, Alpha Centauri, whose distance from our sun is computed as about twenty trillions of miles. Assuming this to be the true distance there could be placed between the two stars fifteen septillions of suns, each with a diameter of 800 thousand miles. Were it possible to expand the earth a million and a quarter times its present size, that is as large as the sun now is, it would then be but one fifteen-septillionth the size necessary to fill the space between the sun and Alpha Centauri. What we know of earth, air, water, rocks, and the metals would not lead us to suppose that these substances could be increased by expansion even a million of times. Could there be such an expansion they would then exist as mere atomic particles of dust incapable of holding heat with the outside element space 300° below zero. Nearly all known substances expand on being heated, though not often to any great extent; as, for example, iron and the metals. But anything that is greatly expanded cools rapidly. Then may we suppose that earth to-day could be expanded into a body large enough to fill the great space it must once have occupied in the state of fire-mist claimed for it? or, if thus expanded, that it would take one year, or even one day, to cool such a body?

We have seen thus how improbable it is that the earth could be expanded to fill the space it must once have occupied according to the nebular theory; and as we imagine the denser any volume is the more it will expand can we suppose the other planets, with a volume thousands of times greater than that of earth but a density not averaging one-fourth as much, will expand to a greater degree? Were they all ground to the finest dust, even like the atoms we detect floating in the sunbeams, they would no more than fill a globe, with the sun for its centre, whose circumference reached out to Neptune.

2d. *Contraction.* It is thought by many that the sun obtains its heat by the contraction of its diameter, and that at the rate of two hundred and twenty feet per year, or four miles a century. Before contraction, then, both the sun and the earth must have been much larger and consequently nearer each other than they are as seen to-day. If the sun's diameter contracts four miles during a century, to increase its size so as to carry it out to Neptune, 3,000 millions of miles distant, would take 1500 million centuries. But that the sun thus receives its heat is a supposition; for how can any one tell that it contracts each century two miles on its radius, when a second represents four hundred and fifty miles, and two miles would be but one-two hundred and fiftieth part of a second?

Should the earth be cooling by expending more heat than it receives, as some claim, it should contract from the loss of heat as well as the sun. But if earth does thus contract it must be smaller than formerly, the sun must have less hold upon it, and with a varying gravitation, must lose its delicate balance.^[2] Yet what proof have we that earth is to-day smaller than it was two thousand years ago?

Further, we find that the more a body contracts the faster it revolves. The sun now revolves in twenty-five days, but when eight million times larger and extended out as far as earth, it must have revolved very slowly; hence with a slow revolution, and at the same time having only four cubic rods of hard substance out of every thirty-three millions of cubic rods, or one cubic mile,—for earth has contracted to one eight-millionth part of the size it then was,—why did not the rocky substance settle to the sun's centre instead of being thrown off to form earth, especially as the sun's gravitation was so great at its surface?

Professor Ball tells us that in gaseous bodies the loss of heat involves a corresponding contraction of the volume, attended with a rise of temperature. To quote his words: "As the temperature of the mass increases the rate at which it parts with heat also increases. The contraction of the volume will proceed at an accelerated pace, and the temperature rise with increasing rapidity. Though the temperature of the gas may at first have been extremely low it will gradually rise until it becomes sufficiently high to render the gas visible by actual incandescence. As the process advances still further the body may pass from a mere nebula into a star-like object. With increase of contraction the pressure also increases and materials which

were originally gaseous will assume more and more a density resembling that of solid bodies.” He says further that should the sun contract into a globe less its present size by one ten-thousandth part of its diameter it would amount to a shrinkage in its diameter of 87 miles. “But,” he continues, “on so mighty a globe this alteration is relatively insignificant; indeed no measurements that could be made at our observatories would be sufficiently delicate to detect a change of this magnitude. Helmholtz has, however, shown that if the sun were to undergo even this small diminution of volume the quantity of heat that would be thereby liberated for the purposes of radiation would supply the sun’s current rate of expenditure for nearly 2000 years. We have no means of knowing at present whether the actual contraction of the sun takes place at this rate or any other rate.”^[3]

Thus we see astronomers admit that a contraction of merely four miles of the sun’s diameter would be sufficient to supply its heat for a century, while a contraction of 87 miles, or $\frac{1}{10,000}$ part of its diameter, would give to it heat for twenty centuries, were the sun gaseous. This being the case how is it possible to detect in this century, with a contraction of but four miles, whether the sun is growing either larger or smaller, or in any wise changing its volume?

When its diameter was twice as large as now it must have been so much cooler that it moved more slowly and radiated less heat. With a diameter of 10 millions of miles, of 100, 1,000, 3,000, or 6,000 millions of miles even, and the sun then more years than it is now days in turning, can we suppose that it revolved swiftly enough to throw off rings; or, with a surface so expanded, was possessed of heat to any great amount? These are thoughts that should be carefully considered in looking at the theory of the formation of worlds from nebulae; for any explanations concerning the existence of a fire-mist so extensive as to reach Neptune’s bounds are of no small consideration, and should be open to careful scrutiny before absolute acceptance.



FOOTNOTES:

[1] Warren. "Recreations in Astronomy," p. 182.

[2] "Laplace has given us proof that the period of the earth's axial rotation has not changed 1-100 of a second of time in two thousand years."

Warren. "Recreations in Astronomy," p. 145.

[3] Ball. "In Starry Realms," p. 31.

CHAPTER II.

DENSITY AND GRAVITATION.

1st. *Density.* The sun's density is one-fourth that of earth, while Mercury's is one-fifth greater than earth's, showing that Mercury's substance must be more than five times denser than the sun's, whereas it is not 36,000,000 miles distant from it; so near, in fact, and so recently thrown off from the sun as to be thought that human beings could not live upon it. Why is it that this dense substance did not, while a portion of the sun, sink to its centre?

It might be said that on cooling, after being thrown from the sun, the body became more and more dense, the same as is said of earth. But if so why is it that Neptune, seventy-five times farther from the sun, has a density only one-fifth that of earth, and Uranus but a little more; while Jupiter's density is less than the sun's, and Saturn's not even one-seventh that of earth? All of these planets lie in an immensely cold and far away region and were thrown off by the sun, if at all, many years before Mercury, and, according to the supposed theory, should be cold bodies.

Some think, with apparent show of reason, that before planet-making began the heavier materials of the general mass had gravitated toward the centre, while the lighter substances remained near the surface. "If so," to quote the words of Prof. Winchell, "the first planets separated would contain more of the substances which, at temperatures familiar to us, make gases and water. Similarly, the later planets disengaged would acquire a large proportion of the substances which form solid rocks. In the case of the earth we may suppose that the greater part was rock-making material, since the earth's specific gravity is so high; but watery stuff in sufficient amount to provide for oceans and rains, went off with the rock-material, and with these, the lighter stuff for an atmosphere. But in the case of Venus, most of the stuff was rock-material, if not the whole of it; while with Mercury it seems probable that little water-stuff was included. In the opposite direction, Saturn, Uranus, and Neptune must have received a large excess of water and atmospheric stuff. It is rational to suppose that their oceans have

always covered the whole land, as ours does more than half. In fact, these bodies must be composed chiefly of water and atmosphere; as their specific gravities are low as water and cork.”^[4]

Now if this is a good explanation what shall we say of the sun from which these planets were separated? If it grew more and more solid as it contracted until Mercury, nearly ten times denser than Saturn, was thrown off, why is not the sun denser than Mercury? whereas we find it with but one-fourth the density of earth.

When the sun reached out to earth it must have had a diameter of nearly 200 million miles, but having now contracted to a diameter of less than one million of miles, should it not have a density ten times greater than earth's, instead of one so much less? For if the solid parts when out at Neptune began to fall toward the sun's centre, they should have continued to fall until they reached it, or until they had met a density greater than their own. We must remember even at the sun's present surface a body would fall with much greater velocity than on earth's surface because of its greater weight. With the density of our earth more than five times that of water, and twice that of solid rock, all heavy substances must gravitate toward its centre; whereas on the surface of the sun the gravitation is more than 27 times stronger. If then the sun were ever a fire-mist reaching out to earth, it would seem that nothing should have prevented the earth from falling with lightning-like speed to the sun's centre, as its volume was 8,000,000 times larger than now, and even its present volume would hold 900 thousand worlds like ours before it would have a like density.

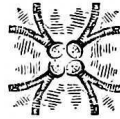
2d. *Gravitation.* We see the sun to-day as a perfect sphere, but does a body that is a sphere ever throw off rings by rotary movement? When a body in its revolutions throws off rings by rotating, instead of being spherical it is of a flattened, or grind-stone shape, and the rings are hurled from it by the centrifugal force overpowering the gravitation; hence we cannot think that the sun's rings,—being of enormous circumference and necessarily of a light or fluid substance in order to be thrown off,—could form into spheres unless the centrifugal force was extremely great.

Let us suppose that Neptune was thrown off from the sun as a ring, like those we see around Saturn; and, as it is now about 3,000 millions of miles distant from the sun's centre, before it was detached it must have had a

diameter of about 6,000 millions of miles, with a circumference of over 18,000 million miles. Now, as Neptune has about 100 times the volume of earth,^[5] its ring could have been no more than 40 miles square; for 1600 square miles multiplied by 18,000 million miles, the distance around that ring, will give more than Neptune's volume. How then could any substance so exceedingly thin draw to itself this enormous distance of 18,000 millions of miles, any more than a thread a thousand miles in length could draw itself together into a ball, without the thread's breaking into a million pieces?

Or, take another theory, and instead of supposing that the sun threw off rings, suppose that its surface cooled and formed into a hard crust. Had the inner sun then shrunk away from the outer could that crust have ever formed into a globe that would rotate around the inner sphere, and if not how could Neptune have been formed? Should such a crust have extended around the sun while spread out to Neptune it must have had a circumference of over 18,000 million miles, as did the ring, with a surface of 108 quintillions of square miles; so the crust could not have exceeded one-sixtieth of an inch in thickness. If, then, Neptune's substance in any way resembles earth's, with a crust of that thickness upon the sun's surface, it must have collapsed in millions of places instead of having broken away from the sun and formed the globe that we now behold. Chambers tells us: "At the surface of the earth a body set free in space falls 16.1 ft. in the first second of time, with a velocity increasing during each succeeding second. A body similarly set free at the surface of the Sun would start with a velocity 27.4 times as great as that of a body falling at the surface of the Earth. This is equivalent to saying that a pound's weight of anything on the Earth would, if removed to the Sun, weigh more than 27 lbs. The centrifugal force, due to the rotation of any body diminishes gravity at its surface. At the Earth's equator the total diminution is $\frac{1}{289}$ part; whilst at the Sun's equator the centrifugal force is only about $\frac{1}{18,000}$ part the force of gravity. It would be necessary that the Sun should turn on its axis 133 times quicker than it does, for the force of gravity to be neutralized. In the case of the Earth, however, a speed of rotation 17 times as great as it is would suffice to produce the same result."^[6]

By this it is seen the centrifugal force is comparatively insignificant with the sun revolving faster than ever before; for, on the principle that the more a body contracts, the swifter it revolves, it must revolve several hundred times faster now than it did when its circumference was at Neptune's bounds. If this be the case it is difficult to believe there was ever a time that the sun could have been larger than at the present, and have had centrifugal force enough to throw off rings from its surface.



FOOTNOTES:

[4] Winchell, "Walks and Talks in the Geological Field," p. 217.

[5] Steele's "New Descriptive Astronomy," p. 174.

[6] Chambers' "Hand Book of Astronomy," p. 6.

CHAPTER III.

THE COOLING OF THE PLANETS.

1st. If the sun was once so much greater that it reached out to Neptune,—as it must have done if Neptune was cast off from it,—would not its poles have flattened like those of Earth, and have cooled first, and any matter thrown off from the globe have been at the poles? Yet the poles of Earth seem scarcely to change as it turns on its axis, and a body thrown off from them would not go around the planet; but, on cooling, fall back to the liquid centre. Such being the case why has earth a density greater than the sun, and what was the power that kept, and still keeps it, from sinking into the sun? Knowing as we do the power of gravitation, we should suppose these masses to be so held that they could not escape from the sun. Saturn, with its three glowing rings, we behold as a blazing star,^[7] but why has it not cooled? The sun must have thrown off the planet millions of years ago, for it now lies about 900 millions of miles distant, and the contracting of the sun on its radius would make it 450 millions of centuries since it spread out as far as Saturn. Hence, if Earth is cold Saturn should be the same in that far-away, colder region; and being ten times farther from the sun than the earth, it should have been formed ten times as long ago, according to the nebular hypothesis.

If Saturn yet throws off rings may not the sun do the same until cool as Saturn and reduced to a like size? At the time the sun spread out to that planet its diameter must have been 1800 millions of miles, whereas now it is about 900,000 miles. With such a diameter shrunken to less than one million of miles we would ask why Saturn is not yet cool? Should we extend the sun's diameter again to reach Saturn, it would have, as stated, a radius of 900 millions of miles; giving 10,000 million times its present volume. By contracting the same at a like rate it would be 450 millions of centuries ago that Saturn was thrown off from the sun. Now, as Saturn has only $\frac{1}{1700}$ part of the sun's volume, with a density much less, and lies far out in space; why in all these millions of years has it not cooled instead of holding heat and continuing to throw off rings? If it possesses the

exceedingly light density sometimes claimed for it,^[8] and has a gravitation at its surface but little more than that of Earth, one would suppose to find it a much colder object than the earth.

[illegible]

Jupiter has still great power over these asteroids, and if so the sun should have a much greater power because of its size and proximity.



FOOTNOTES:

[7] “Saturn has a mean diameter that is about nine times that of Earth while his volume exceeds hers more than 700 times. Within an extreme span of upwards of four millions of miles on either side of Saturn’s globe there circle eight satellites and two millions of worlds, the least of which is probably as large as Mars. Then within the path of the innermost of these moons there is the wonderful ring system of Saturn. The span of this system of rings amounts to about 176,000 miles; that is its outermost edge lies about 88,000 miles from Saturn’s centre, while the complete system has a breadth of 37,600 miles, but the innermost part, to a breadth of nearly 9,000 miles is dark. Through this dark ring the outline of Saturn’s disc can be clearly perceived. In fact this wonderful dark ring is transparent. The bright parts of the system form two rings, separated from each other by a dark, but not perfectly black, circular division about 1700 miles broad.”

Proctor, “Expanse of the Heavens,” p. 96.

[8] “Its mean density is less than that of any known planet, being less than one-seventh of the earth.”

Proctor, “Expanse of the Heavens,” p. 97.

[9] Proctor, “Poetry of Astronomy,” p. 366.

CHAPTER IV.

SPACE.

If all the stars that we now see in space were once fire-mist, there must have been another region equally large filled with some substance of a temperature 600° below zero, in order to equalize the heat of the fire-mist and leave space 300° below zero, as it now is in the vast region that surrounds existing stars.^[10] What must that cold of 600° have been, for what is that of 300° even when water freezes at 32° above zero and mercury at 30° below! Where that cold space was we know not, neither do we know what was in space before the fire-mist.^[11]

If the nebular theory will sufficiently account for the throwing-off and cooling of the planets, will it likewise account for the millions of suns disseminated throughout space? In the first place, let us suppose that all these suns were made at the same time and filled space with the same substance as our earth, only in a diffused state. In such a case I cannot conceive how it could break up and resolve into stars, for there being no space to turn in all would revolve together. The pulp within the rind of an orange could not be cut into circles and caused to revolve inside of that rind; so fire-mist once filling the immensity of space must have continued to revolve in an unbroken mass. Secondly; If the stars were formed at different periods of time should not many be still forming? whereas space appears wondrously clear. We would think if all this great space were to-day filled with nebulae composed of material similar to that of earth, we could not see; for our vision of the heavens would be obstructed thereby, and we should know nothing about the countless numbers of stars that, with the aid of the telescope, now make the very heavens to blaze with light. Why is it that the space about all, or nearly all, of the suns has thus cleared if a state of fire-mist is common to all of them? It is true Nebulae are seen that appear like vast fields of dim light, but they are often resolved into stars when examined with powerful instruments, and the nebulae that cannot be thus resolved into suns occupy but a small portion of the heavens. Less than one cubic foot is now left out of the 147,200 millions of cubic feet in every cubic mile of fire-mist that the sun's sphere must have contained when it

reached out to the planet Neptune, and what has become of it all? If it is not in our system it must have gone into others. Again; if among the stars that we now see there are a million, or even a thousand times as many dark bodies as luminous ones, they must in a measure obstruct the light of the glowing suns. But were this known to be true it would not prove that these bodies were not originally formed dark objects; and who but an omnipotent and omniscient Ruler could prevent repeated collisions among them?

The nebular theory accounts only for our own solar system, and yet all the stars that we see in the heavens can be no less wondrous than our own flaming sun. That we see their light even, at so great a distance, is proof that they can be no less great. We must account, then, not only for our own planetary system, but for the countless millions that exist; and we must concede that the laws that govern one would, presumably, govern every one. Hence, a period of fire-mist for one implies the same for all.

If the distance from our sun to the next nearest sun is 20 trillions of miles, a sphere whose radius reaches half way, that is ten trillion miles, could contain eight sextillions of suns, each with a diameter of one million of miles. Now let a single grass seed represent one of these suns, and twenty-seven millions of them one cubic inch. Should we fill a bin one mile long, broad, and high, with such seeds, there would yet be more suns in the above sphere than it would take of grass seeds to fill this bin holding seven sextillions of them. When we think that each grass seed is to represent a sun two million times larger than our earth,—for each sun with a million of miles diameter would,—the thought of the contents of such a sphere is overwhelming.

Professor Mitchell tells us that with a telescope light from distant nebulae can be seen that has been thirty millions of years on its journey. Let us imagine a sphere with a radius of one million years of light's flight at the rate of twelve million miles per minute, viz., six quintillions of miles. A sphere of such a radius would contain as many spheres of ten trillions of miles radius as one could put cubic inches in 850 cubic miles, or 216,000 trillions. This sphere could hold as many suns of one million miles diameter as there might be grass seeds that would cover a million of earth's for one mile in depth, each earth containing on its surface 200 millions of square miles, and every cubic inch thereof representing 27 millions of grass seeds. When we think that each seed represents two millions of earths can we

comprehend the greatness of the universe? Remember the same calculations that enable us to determine the number of cubic feet in our earth, sun, or any other sphere, bring these astounding but certain results. And when we further remember this space is every where so clear and powerful that it transmits light at the rate of over eleven million miles each minute, enabling us to see in every part of the heavens the wonderful stars at an immense distance; and that if we bring to our aid the most powerful telescopes they only tend to magnify the already vast number of stars; we are led to realize that beyond all things else we can conceive space is the most astonishing and wonderful, excepting its great Creator. While some think that the sun is cooling and all its planets will be affected thereby, and refer to the destruction and instability of earthly things; where is there a single atom of this inconceivably vast, vibratory space that reveals the least change or destructiveness, although it must have existed for millions of ages.

Sir Robert S. Ball has given a very marked illustration of the wonderful magnitude of space in his late book entitled "In the High Heavens," from which we quote as follows: "Summon up to your imagination the most distant star that can be seen with the unaided eye. Then think of the minutest star that our most potent telescope can disclose. Think of the tiniest stellar point of light which could possibly be depicted on the most sensitive photographic plate after hours of exposure to the heavens. Think, indeed, of the very remotest star which, by any conceivable device, can be rendered perceptible to our consciousness. Doubtless that star is thousands of billions of miles from earth; doubtless the light from it requires thousands of years, and some astronomers have said millions of years, to span the abyss which intervenes between our globe and those distant regions. But, nevertheless, there is a certain number of miles, even though we know it not, at which the remotest stars known to us must lie. I do not speak of the most distant star which the universe may possibly contain; I only refer to the most distant star that we can possibly bring within our ken.

Imagine a great sphere to be described with its centre at our earth, and with a radius extending all the way from the earth to this last star knowable by man. Every star that we can see, every star whose existence becomes disclosed to us on our photographs, lies inside this sphere; as to the orbs which may lie outside that sphere we can know nothing by direct observation. The imagination doubtless suggests with irresistible emphasis,

that this outer region is also occupied by stars and nebulae, suns and worlds, in the same manner as the interior of that mighty sphere whose contents are more or less accessible to our scrutiny. It would do utter violence to our notions of the law of continuity to assume that all the existent matter in the universe happened to lie inside this sphere; we need only mention such a supposition to dismiss it as wholly indefensible. I do not now make any attempt to express the number of miles in the diameter of the sphere which limits the extent of space known directly to man. What that number may be is quite immaterial for our present purpose. But the point that I especially want to bring out is that the volume occupied by this stupendous globe, which includes within it all possible visible material, must be but a speck when compared with the space which contains it. Think of the water in the Atlantic Ocean, and think of the water in a single drop. As the drop is to the Atlantic Ocean so is the sphere which we have been trying to conceive to the boundless extent of space. As far as we know it would seem that there could be quite as many of such spheres in space as there are drops of water in the Atlantic Ocean.”

Now with this defining of space it is evident that it matters but little what the material substances of the universe may be. If the hundreds of millions of bright suns—which are thought to be few in proportion to the invisible, dark worlds scattered throughout space—are but as a single drop of water to all the Atlantic Ocean; then we are compelled to admit that our earth and sun, and even the great host of luminous orbs, must be of little consequence beside this infinitely vaster and more intensely active ether.

It behooves us above all things, then, to inquire what this amazingly great space may be. Let us quote further from the above author: “Every particle of matter whether solid, liquid, or gaseous, is composed of molecules. No doubt these molecules are so numerous that even in the air we breathe the capacity of a lady’s thimble would contain a multitude of molecules so great that it has to be enumerated by billions.” Again, “The air is ultimately composed of myriads of separate particles. Each of these little particles, no matter how quiet the air as a whole may seem, is in a state of intensely rapid movement. Picture to yourself incalculable myriads of little objects, each dashing about with a speed as great as that of a rifle bullet, and often indeed far greater. The little particles are so minute that it would take about fifty millions of them, placed side by side, to extend over a single inch. The

smallest object which we can discern with a microscope is perhaps one hundred-thousandth of an inch in length. The little gaseous molecule would therefore require to possess a diameter about five hundred times greater than that which it actually has if it were to be large enough to admit of inspection by the utmost microscopic powers which we can bring to bear upon it. And yet, notwithstanding the fact that these particles are so extremely minute, we are able to reason about their existence, to discover many of their properties, and to ascertain the laws of their action in such a way as to throw light into many obscure places of nature. I do not, indeed, know any doctrine in modern science of a more instructive character than that which teaches us the composition of gases." If this be true of air what then can be said of space, or ether, in which all worlds float as easily as the motes in a sunbeam; that space which transmits light everywhere with a precision that never varies? And what is light, indeed, but vibrations of ether from 400 trillions to 800 trillions per second, giving all the colors from violet to red? There is not an atom of ether in all space, so far as science can detect, that has ever ceased to vibrate, or ever will, with the startling rapidity above expressed.

But, to further illustrate the magnitude of space, let us again take the flight of light as a basis for our calculations. As there are 525,600 minutes in a year, light—moving at the rate of eleven millions of miles each minute—must travel in one year 5,781,600 millions of miles. With that number of miles as radius of a sphere, of which earth is the centre, the diameter will be 11,563,200 millions of miles, and the surface of the sphere 401,112,000 quintillions of square miles, while it will possess a volume of about 800 undecillions of cubic miles. Dividing this number by the 260,000 millions of cubic miles that earth contains we have 3,000 septillions, the number of earths that such a sphere could contain. Now in an ocean 5,000 miles long, 3,000 miles broad, and 3 miles deep there will be 45 millions of cubic miles, or 250 trillions of inches. Allowing 200 drops of water to each cubic inch, we have in one cubic mile 50 quadrillions of drops, and in that ocean 2,250 sextillions of drops of water. Dividing 3,000 septillions—the number of earths in the above sphere—by the number of drops of water in the ocean, we find we would need 1300 such oceans to furnish enough drops of water to equal the number of earths that could be placed in a sphere whose radius is but the number of miles that light travels in a single year. With a radius equal to one hundred years of light's flight a sphere might contain as

many earths as there would be drops of water in 1300 millions of such oceans; while a radius of light's flight for 100,000 years could hold as many earths as there were drops of water in 1300 quadrillions of such oceans, or a number of oceans equalling the drops of water in 26 cubic miles. Again, in a *million* years of light's flight there might be as many earths as drops of water in 1300 quintillions of such oceans, or the number of oceans equalling the drops of water in 26,000 cubic miles.

If a sphere with a radius of light's flight for but one year could contain 3,000 septillions of bodies like our earth, and yet that sphere be but an atom in space, it would seem that space might be *infinite* in its extent, with our conception of infinity. But if space is *finite* and light, after a flight of a million of millions of years, reaches its utmost bound; then that light, if still existing and radiating 400 trillions of vibrations each second, can double that time and return. For if not cooled in a million of millions of years the supposition is that it will not cool in twice that time.

Still, again, let us conceive of a sphere, but this time with a radius—not of *ten* millions of years of light's flight, which would contain as many earths as drops in 1300 quintillions of such oceans, a number equalling more than half the drops in an ocean that contained 45 millions of cubic miles—but we will take less than ten minutes of light's flight with a radius of 100 millions of miles, which is a little more than the distance from the earth to the sun. We find that this sphere would contain 15 trillions of earths; a number ten thousand times greater than that of all the people living upon earth. When we think that each one of those earths would contain 260,000 millions of cubic miles it is seemingly all that our minds can well grasp.

We may not know how many millions of years each atom of ether has been in existence, but probably before any worlds ever floated within it; and how can we conceive the thought of death in this immensity of space where there is not the slightest indication of subsidence or decay? We must remember that this ether is a million times more active than air and possesses energies that we cannot conceive of in the more solid substances of the spheres. Then, whether the nebular hypothesis of the cooling of the sun and worlds be right or wrong, we may not detect it from any evidence that space gives thereof; and we can but believe that, if it be a fact, it must be so only because He who formed this wondrous space has in their destiny

an object. The Eternal Presence may give to this vast, eternal space a glow that needs not the light of sun, moon, or the shining host of stars, even though all these mighty orbs that we now behold should be plunged into everlasting darkness.

In order to better comprehend the mighty vastness of this space we will measure it again into units of 260,000 millions of cubic miles, (the size of earth), and with a radius based on the number of miles of light's flight in minutes, days, months, and years, ascertain the number of different sized spheres that such radii would form. A sphere with earth as its centre, and a radius of one minute of light's flight, would contain 21 billions of earths; a sphere whose radius was *nine* minutes would contain 15,310 billions of earths, one of an hour's radius 4,500 trillions, one of a day's 62 quintillions, a week's 20 sextillions, a month's 1675 sextillions, and a year's radius of the flight of light three octillions. To go still further; 10 year's radius of light's flight would contain three nonillions, 100 year's three decillions, 1,000 year's three undecillions, 1,000,000 year's three quattuordecillions, and a thousand million years radius would contain three septendecillions of such earths. Now to compare these great numbers we will suppose, as before, that there are 200 drops of water in one cubic inch, making 50 quadrillions of drops in one cubic mile of water, 50 quintillions in one thousand cubic miles, 50 sextillions in one million of such miles, 50 septillions in one thousand millions, and 50 octillions in one million millions of cubic miles.

We need go no further for the immensity of space is seen by comparison of the earths in the above spheres with the number of drops of water in all those cubic miles. We find the sphere that has but one day's flight of light as a radius will contain more earths of 260,000 millions of cubic miles each, than there are water-drops in 1,000 cubic miles; or, in other words, as many drops as there would be in 100 Sounds like Long Island from Bridgeport to New York, allowing those sounds to average 100 feet in depth and each contain ten cubic miles of water.

The thought so astounding and well nigh inconceivable is that a sphere of such vast dimensions has only a radius of a single day's flight of light, light that starting from its centre would pass the circumference of the sphere in one day; while a sphere with a radius of *ten* year's flight of light—a distance only a little way beyond the nearest known star in the northern

heavens, 61 Cygni—could contain as many earths as there are drops in 60 millions of millions of cubic miles. Yet a sphere with that radius of ten year's flight would cover such an enormous quantity of space as to be beyond our comprehension and still be of no consequence in comparison with the spheres of one hundred, one thousand, one million, or one thousand millions of years radii.

Why I thus compute 1,000 millions of years of light's radius is because it is possible with powerful telescopes to detect light coming that distance; and when we contemplate that our nearest star, Alpha Centauri, is twenty trillions of miles distant, and 61 Cygni more than twice as far,—or *seven* year's flight of light,—and that the Polar star is some fifty years of light's flight removed from us; then, if all suns are no nearer to each other than these are to our sun, we are led to believe we may see light with our unaided vision that has been on its journey for a million years. Some detect with the naked eye the light that comes from the nebula in the Sword's-Handle of Orion, which is thought to contain two trillions, two hundred billions of stars; and if all these stars are no nearer to each other than Alpha Centauri is to our sun, it cannot be otherwise than that some of them are so distant their light may have been travelling for a million years before it comes to our sight. If this be true, with telescopes that penetrate a thousand times farther into the heavens, we may possibly see light that has been on its voyage to earth one thousand millions of years. But the number of earths a sphere of 1,000 millions of years of light's flight would contain makes the number of drops of water in sixty millions of millions of cubic miles (equalling but the earths in a sphere of *ten* years radius) sink into utter insignificance. And yet a sphere of this dimension, even, may be small beside the whole universe of ether whose every atom in all the millions of years past, as far as known, has not in the least diminished its wonderful energy; and, with lightning-like transmission, still brings to our sight light from distant stars, all of which are so distant that they do not materially increase the present flow of light. For if but a single one were to come between us and the sun the increase of that light and heat would be unendurable.

Finally, as space thus surpasses everything in its greatness excepting its Creator, let us contemplate it once again in the following manner. Suppose we could create 40,000 millions of cubic miles of ether daily; we would at

the end of the week have an amount equalling the volume of our earth. If we continued that creation daily,—not for hundreds, thousands, or millions of years even,—but for 1,000 millions of millions of years, we should then have as many such volumes of ether the size of earth as there are drops of water in one cubic mile. But what of all this? Admitting that we could create bodies as fast as our earth was made, and continue to do so for 1,000 millions of millions of years, it would still be of little account; for its insignificance is seen when we remember it is but one-fifty-eight thousand millionth part of what a sphere would contain with a radius of one year's flight of light. To think of 40,000 millions of cubic miles being created daily, with the process continued for 1,000 millions of millions of years, and *that* vast quantity still further expanded 58,000 millions of times, would seem the extent of greatness itself and beyond all human realization. Yet we can still say all this would be of slight consequence, for it represents a sphere whose radius is but one year of light's flight, and that sphere compared to one of 1,000 millions of years radius is no more than a boy's marble,—with a radius of but one-fourth of an inch,—to earth whose radius is 4,000 miles, and a thousand million times larger than the radius of the marble.

We have endeavored by the above comparisons to show the vastness of space; and although our efforts must be in vain because of our inability to comprehend such great figures, still it is pleasant to contemplate this ether, upon which time has so little effect, because man is mortal and the 1500 millions of people that pass from earth every 33 years makes it seem to him that death reigns everywhere. But when we consider that the only things essentially effected by time are the animal and vegetable kingdom, and that all men now living could occupy about one-fifteenth of one cubic mile of earth; we see that death, even in the animal kingdom, is confined to a few cubic miles, and to but a few more in the vegetable kingdom. The inconceivable millions of millions of cubic miles in a sphere with a radius of one million years of light's flight is in no manner affected, as we perceive, by the dread power that we recognize in the dissolution of mortal bodies.

Even these bodies, as we understand, are changed only in the combination of molecules, and by that change the immortal spirit, that is as imperceptible as the wind, of which we are told we cannot tell “whence it

cometh nor whither it goeth,” is set free; the Bible teaching us its eternal destiny. The greatest and best things that men here possess are love and goodness, and should not the Almighty possess these virtues beyond any of His creatures? Possessing these how could He otherwise than send to Earth His Son for their salvation from sin; for we find men here, even, who are willing to peril their lives for the salvation of their fellows, and would undoubtedly be kind to an inferior animal, though an ant or worm, had they created it and knew it loved and worshiped them. Akin to this, we believe, is man’s relation to his Creator and the Creator of this mighty and seemingly everlasting Space.



FOOTNOTES:

[10] For example: Take two rooms and heat one as hot as the sun now is what must the cold in the next room be to equalize the heat to 300° below zero?

[11] We do not inquire how the original nebula came into being; our history must commence with the actual existence of this nebula. There is, let it be confessed, a great deal of obscurity still clinging to the subject. Though we may be sure, that the great nebula once existed we cannot with much confidence trace out the method by which the planets were actually formed.

Sir Robert Ball, “In Starry Realms,” p. 348.

CHAPTER V.

THE EARTH'S CRUST.

Were earth's crust no thicker in comparison than the skin of an apple or peach, or $\frac{1}{100}$ of its radius even, still man has never penetrated to one-twentieth of that depth; for I know of no place where he has as yet reached one mile below earth's general level, unless it be in ocean-soundings. In most places the mines or gorges have been in the mountains high above the valleys, consequently we know little of what is far beneath the earth's surface. How could we judge of the inside of nuts, fruits, grains, or vegetables if we had penetrated to but one-twentieth of their coverings?

Take, for instance, a chestnut that all boys are familiar with, and see what we may learn from it. We find that it is covered with a tough, brown shell which in turn is enclosed in a large prickly burr ten times the size of the nut. This protects it until ripened and then opens to let the nut fall. The burr is fastened to a large forest tree whose roots are deep in the ground; a tree that had been growing for years before the chestnut was produced, and first started from another nut of its kind. From the time its growth began there was nothing in root, trunk, limb, or leaves for years that in any way resembled a chestnut; nor in the blossom or burr even, until the nut ripened and fell to the ground. Could one think, who had never before seen the nut, that from such a tree a chestnut could be produced?

The same is true of all manner of nuts, fruits, and vegetables. It is even true of animals, for they are enclosed in such varied coverings that their nature is often quite concealed from us. We may as well believe there is something in earth's interior, away from winds and cold, as precious over and above its crust as the chestnut is better than the burr that encloses it; the fruit better than its rind; or fish, birds, and animals better than their skins. Knowing this, man, while he would grasp the greatness of the universe, may somewhat under-rate the contents of his own earth.

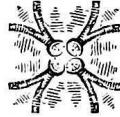
Dr. Winchell in speaking of cold has said; "It has been demonstrated that an ice-cap resting several thousand years over any considerable portion of the surface would so reduce the subjacent temperature of the earth that for

many centuries after the disappearance of the ice, a decrease of temperature would be discovered in penetrating downward. Even centuries later, so much cold would still remain within the earth, that the rate of increase of temperature would be less than if the ice-cap had not existed; and after 3,600 years, that rate would be only half the normal rate.”^[12]

Now if the earth’s crust will thus retain the cold why may it not as well retain the heat? In that case any excess of heat escaping from earth, over and above the heat it constantly receives from the sun, may be readily accounted for. It is claimed that the earth has large quantities of sunlight and heat stored in a liquid state, as petroleum; in a gaseous state, as natural gas; and in a solidified form, as coal; some of which are found at a depth of 1600 or more feet, although coal is sometimes found near the surface, and even upon the mountains. If these are all stored sunlight, why do they not accumulate over earth’s entire surface as well as in certain localities,—especially in the torrid zone where great heat exists,—and accumulate to-day as well as in former ages?

When we remember that the Mesozoic aeon was preceded by the long Palæozoic, and earlier by the Eozoic,—whose aeon has been laid down at eleven million years, with a strata at least 50,000 feet thick of hardest rock,—what could have been the cause of the submerging of the lands where these gases and coal lay? For millions of years the Eozoic strata had upheld the oceans that were two or more miles in depth, and of enormous weight. If the earth was cooling all those years the crust must have been hardening, and what added weight could have been at so late a period to cause the submerging of those lands? It has been thought that in the glacial period when the accumulation of ice over a portion of earth’s surface lay 5,000 feet thick that its weight might have depressed the terrestrial crust.^[13] But ice being lighter than water, even though it were of that thickness, would not have had the weight of water a mile in depth; whereas more than twice that depth of water rested upon three-fourths of the globe. Moreover the ice-period was many millions of years later than the Palæozoic and Eozoic ages, and had the earth all this time been growing colder its crust must have been thickening, making it capable of enduring almost any pressure. Any crust, as ice upon water or the shell upon an egg, will bear more than its whole weight before it will sink into the substance upon which it rests.

It might be thought that earth, while cooling, would shrink beneath its crust and leave a vacuum, as ice sometimes does upon a small surface. This could not happen as the vast surface of extended crust, by the enormous pressure of air upon it, would be held closely pressed to whatever was beneath. For, although earth's surface is convex, it is still 25,000 miles around it, a distance so great that its crust would have almost the same pressure as if the surface were level.



FOOTNOTES:

[12] Winchell, "Walks and Talks in the Geological Field," p. 99.

[13] Winchell, "Walks and Talks in the Geological Field," p. 275.

CHAPTER VI.

THE EARTH'S HEAT.

That the heat of the earth increases as we penetrate its surface has been learned from mining, tunneling, and the boring of wells. Yet on testing the wells they do not seem to show any uniform temperature. The deepest Artesian well is in St. Louis and has a depth of 3,843½ feet, while its water is found to be of a temperature of 105°. ^[14] This would seem to indicate that the earth is hotter as we descend into it; still, there may be reasonable causes therefore without its heat extending after all to any great depth. When we remember that the earth is 4,000 miles from surface to centre we find that this deep well is not even $\frac{1}{5000}$ part of the distance, and what may be in the interior of the earth is yet quite uncertain. It may possess elements, that from earth's swift revolutions on its axis, and far swifter flight through space, would supply any loss of heat over and above what is received from the sun. Heat may also penetrate earth's surface more easily than it escapes, for earth is surrounded by an atmosphere that receives the sunlight readily, but not so readily lets it go; and prevents the outside cold of 200° below zero from falling upon it nightly.

Further it is said, with seeming reasonableness; "No rock has the requisite rigidity to resist the crushing weight of a mountain twenty miles high." Whatever movements may take place in the earth's crust, involve masses so great and forces so enormous that the resistances of ordinary matter are inconsiderable. The most solid rocks are essentially fluid or viscid. Now, such movements must necessarily result from two causes: First, a slow shrinkage of the earth through loss of heat; secondly, the attraction of the sun and moon, which cause tidal protuberances on the surface of the earth, however rigid it may be; and these, continually shifting their positions, as the oceanic tides do, result in daily motions adequate to develop a large amount of frictional heat. ^[15]

This last occasioning of heat we would especially notice, and, perhaps, amply account for the present known heat. If at the bottom of said Artesian well, two-thirds of a mile deep, the temperature is 105°, why is it at the

bottom of the ocean, five miles toward earth's centre, that the water is ice cold? It is admitted that the question concerning internal heat is imperfectly understood. "We neither know," says Professor Winchell, "at what depth it exists, at what ratio it increases, nor what is its cause or source. Nor do we know whether the deep interior is in a solid or a liquid state. Assuming the rate of increase to be one degree for 60 feet of descent, we should obtain, in the latitude of New York, heat enough to boil water at a depth of about 9,000 feet." We note that in proportion to the depth of the well it should have a temperature of at least 500°, or a heat that would cause the waters of the ocean overlying three-fourths of earth's surface to boil; especially in deep waters where there is but little of earth's sediment, and where its crust must be necessarily thinner than the elevated land.

Again; We are told that the moon is scarred all over with volcanic craters, some of which are 100 miles in diameter; but what volcanic crater on earth could be detected 240,000 miles away by any telescope that magnified 1,000 times, or even be seen by the naked eye at a distance of 240 miles? There are, to be sure, at the present day volcanoes that give evidence of great internal heat, as Etna and Vesuvius. These compared to earth's vast surface of 200 millions of square miles, and vaster volume, would be no more than a burning leaf in a forest of trees. The many extinct ones show it must have been the same in past ages, but how soon these are seized upon by vegetation and hidden from view! We are told that masses of lava are very poor conductors of heat and have been found burning a century after their eruption.^[16] This being the case how long might heat be imprisoned in earth, when it has been stated that cold might not all escape in 7,000 years? May not this imprisoned heat be the source of the escaping of any excessive heat over and above that which earth has received; and all these volcanoes—and the hot springs as well—be caused by smouldering under-ground fires of liquid, gaseous, or solidified heat; generated, we know not how, from either the sun's or the earth's heat? For if the sun's heat can be stored in cold bodies to be used thousands, or even millions of years afterward, may not earth's escaping heat be likewise returned to earth, and every particle of it be held by earth, as water and air are held by it?

The petroleum, natural gas, and coal that have been little known until the last fifty years, are all lying within one half mile of earth's surface and of sufficient quantity to make several volcanoes, could oxygen be brought to

fan them into a flame. What materials equally as combustible may be discovered when the earth is penetrated in other places, and for another half mile below its surface, we do not know; neither do we know what may be revealed as we descend deeper and deeper into earth's every portion until reaching a depth of an hundred or a thousand miles. But who shall say there are not as wondrous things yet to come from beneath earth's crust as have ever been found upon, or near, its surface? Man has already discovered over sixty elements some of which, were they in abundance, would give a very enduring flame. When we see chemists separating water for burning, or consuming steel files in combustion we may be prepared for other startling discoveries. What we already know, through cyclones and hurricanes, concerning the power of air, that is seemingly so subtle and still, should lead us to believe that we have as yet little understanding relative to the hidden truths of earth; and what its interior may possess of heat and other elements is not for us to say with any degree of certainty.

The thought we may now well entertain is this; it is remarkable that the earth's surface where we are dwelling is well adapted for our existence. At present there is life here; that death was in the past, and destruction will be in the future, we may believe as we choose, but have no certainty of the fact; for the indestructibleness of every atom of ether in the universe would seemingly question the power of time to work destruction to our earth so minute in comparison.



FOOTNOTES:

[14] Winchell, "Walks and Talks in the Geological Field," p. 98.

[15] Winchell, "Walks and Talks in the Geological Field," p. 101.

[16] Winchell, "Walks and Talks in the Geological Field," p. 99.

CHAPTER VII.

THE SUN'S LIGHT AND HEAT.

I conceive that one reason why scientists believe in the nebular hypothesis is because of their knowing that heat consumes. One cannot contemplate a burning object without perceiving that it grows smaller and smaller; therefore why should not the sun with its flaming hydrogen, rising sometimes to a height of 200,000 miles, consume the sun? It is claimed but $\frac{1}{2300}$ millionth of its force reaches earth, and yet it is asserted that the sun could melt 287,200,000 cubic miles of ice per second without quenching its heat.^[17] From what we know appertaining to heat, how can we think it is otherwise than reducing the sun's volume?

Yet we must remember there are other things as difficult of comprehension. We see the mist rising from the ocean and forming into clouds that drift through the heavens when moved by the winds, and we might well believe in future years the oceans will be drained of their contents. But when we learn that all those waters pass into the sky but to condense and fall to earth, and then through streams again reach the oceans, we can readily understand that they may be the same to-day as when created; nor conceive how it will be otherwise to the end of time.

Again, when a boiler of water seems wasting through invisible steam, expanded 1800 times its original bulk, we might well believe it is being destroyed only we have learned all that steam in some manner cools and forms again the first element H^2O , not one particle being lost. When any body is burned and we see the flames ascending into the air, we say it is being destroyed, for so it seems; but chemists tell us the form alone is changed and the weight after burning identical with its first weight; showing thereby that not one particle of earth can either be formed or destroyed, but simply changed from combinations of molecules to simple molecules, or *vice-versa*.

We see earth everywhere surrounded by an atmospheric sea, not of oxygen and hydrogen, but of oxygen and nitrogen—four-fifths being

nitrogen—and were the other fifth the same no life could exist in it. Scientists tell us this atmosphere extends from one to two hundred miles into the sky, but is densest at the earth's surface; and as one ascends rapidly rarifies so that at a height of a few miles no life can exist. From what is known of winds and cyclones one might expect the atmosphere would be torn from earth, especially as earth moves at the rate of 1100 miles per minute, and revolves on its axis about the same number of miles per hour. One would suppose, at least, that which is highest and thinnest must be left behind in space; yet we cannot learn since Earth's creation that any of it has been thus lost.

Furthermore, when we contemplate earth's delicate poise of forces,—"No balance turning to $\frac{1}{1000}$ of a grain being more delicate,"—we may well believe the sun is the same to-day as two thousand years ago; for it could not have wasted any of its substance without having thereby affected the gravitation of the earth. In some manner, like the waters that rise from ocean and return again; like the steam, and burning bodies that are not lost; and like the atmosphere the earth holds, the sun may be also holding its every atom of heat; though changed, perhaps, in some of its combinations. We must remember that the sun's rays, as they pass into space, can be seen by us only when the vibrations are between 400 and 800 trillions per seconds, for above or below that number they are invisible to human sight. Were these rays condensed, or in some manner changed, making as great a difference between them as there is difference between the vapors rising from the ocean to the clouds, and the streams returning to them; they might before their condensation, or change, give different vibrations from the ones they would afterward give. While the heat escaping from one gave light the others returning might be invisible to us. It is evident that were the sun a dark object it would be invisible to us, as would the moon without the sun's rays resting upon it.

May not all light be restored to the sun, and thus keep up its supply of heat—as well as vapors be returned to ocean—instead of supposing it is caused by contraction? The only sounds that our ears can detect are vibrations between 16.5 and 38,000 per second and unquestionably all above 38,000, even up to and above those of light, would give sound had we the faculties to detect it. Had we then the right senses all vibrations

below 400 trillions or above 800 trillions per second would be visible as well as the vibrations that give us light.

Though it is difficult to understand, from all the materials that we are familiar with, how a fire can burn without its substance being consumed, we should remember the bush that astonished Moses by burning without consuming. As little can we conceive that there is a great globe of fire keeping up its flame and heat for thousands of years without being diminished. The sun, although seeming to us small as a ball, is visible at a distance of over 90,000,000 miles. Could we conceive of any object, even though a million times larger than earth, being seen by us at that distance only from the fact that it is thus daily seen, and apparently is the same size as when our eyes first rested upon it. Even these visible truths are beyond our conception, and knowing that it is so we should feel that any truth, however astonishingly great, may be possible.

When we realize that without the sun's heat and light we ourselves could not here exist; that it has power to lift the waters that are unfit for man's use and restore them again in a purified state; and power to produce food for him both in the animal and vegetable kingdom, the truth is not lessened. Were the sun nearer the earth or farther away all life would here be destroyed; and should any one of all the suns in space vary the least in its orbit it would be the destruction of earth. Yet as far as we have learned no sun has ever come within twenty trillions of miles to interfere with our globe, and from all the above facts we may well believe there are truths concerning the sun's light and heat that we do not yet understand.

That the ocean keeps up its supply of radiation, and as far as we see never diminishes; that the heat from volcanoes and all other fires of earth does not escape from earth's hold, but is returned in new forms to be used again; leads us to question whether earth's heat passes beyond its own atmosphere. The heat of our great solar light may likewise be equal to that of Adam's day and continue thus unwasted until the end of time, or until its great Author sees fit to change it.

While astronomers tell us that the diminution of the sun's diameter $\frac{1}{10,000}$ part would liberate heat enough to supply its current expenditure for about 2,000 years, they have also shown that it could be supplied by the friction meteors would cause by rushing into the sun; provided that the

number falling into it in one year equalled the moon's volume.^[18] We can know little of the number there may be, for only that they occasionally fall into our atmosphere and are instantly burned, we should not know of their existence. But it is said that every 33 years we pass through a shoal of them 100,000 miles broad, and many thousand times greater in length, and that it has been thus for centuries. Prof. Newton estimates the average number of meteors that traverse our atmosphere daily, large enough to be visible to the eye on a dark night, is 7,500,000. With the telescope-meteors added, the number is increased to 400 millions. As the sun is more than a million times the size of the earth should the number falling into it be increased at the same rate it might reach 150 quadrillions daily, 170,000 falling on every square mile of the sun's surface.

Again; it is stated by Sir Robert Ball that a body of a pound's weight falling from a great distance into the sun, might, in the course of its friction through the sun's atmosphere, generate as much heat as would be produced by the combustion of many times its own weight of coal, if consumed under the most favorable circumstances. Is there sufficient evidence yet given to prove that this is not a source of the sun's light and heat instead of contraction? The moon, we will say, contains 5,000 millions of cubic miles. If the sun's radius was 100 millions of miles, or extended as far as earth, it would have a surface of 120,000 trillions of square miles. Place the earth on its surface and it would occupy less than 1,000 millionth of that surface, allowing it to settle into it one-half. The moon's volume if spread over earth's surface would cover it but twenty-five miles deep. If then, the same quantity of material were spread over the sun while reaching out to earth it would be covered by it less than $\frac{1}{300}$ of an inch in depth; so thinly, in fact that an apple-skin would be thick in comparison. Are we prepared to say this amount does not actually accumulate year by year on the surface of earth, for we are told: "The world is thus pelted on all sides day and night, year after year, century after century, by troops and battalions of shooting stars of every size, from objects not much larger than grains of sand up to mighty masses which can only be expressed in tons. In the lapse of ages our globe must thus be gradually growing by the everlasting deposit of meteoric debris. Looking back through the vista of time past, it becomes impossible to estimate how much of the solid earth may not owe its origin to this celestial source."^[19]

But as the sun does not extend out to earth let us see how deeply the moon's volume would cover a globe one million miles in diameter having a surface of three trillions of square miles. Spread the moon's volume upon this and we find it would cover the sun about nine feet deep. This being for one year it would be only at the rate of nine inches per month, or one-third of an inch per day. Snow would cover earth to that depth in half an hour, while a mist or dew could cover it in about twenty-four hours. Thus a constant deposit of meteoric dust even like dew would give to the sun a volume equal to the moon's in about one year. With these facts before us let us notice what is actually observed about the sun's corona, so plainly seen when the sun is totally eclipsed; for can that corona be less than the dew that falls upon earth if it is thus visible at a distance of more than ninety millions of miles? We are told, "The corona is a vast shell of unknown vapors in a highly attenuated state many thousands of miles thick, and observed to extend at least one-half a degree from what is ordinarily taken to be the visible edge of the sun." Is it, then, too much to believe it is helping to keep up the light and heat of the sun? It is further asserted that its depth is nearly 100,000 miles and "consists of reflected light, sent to us from dust particles or meteoroids giving new densities and rarities that cause the changeful light. Whether they are there by constant projection, and fall again to the sun, or are held by electric influence, or by force of orbital revolution, we do not know."^[20]

The same author quotes from Professor Pierce: "The heat which the earth receives directly from meteors is the same in amount which it receives from the sun by radiation, and that the sun receives five-sixths of its heat from the meteors that fall upon it." Prof. Langley has stated that no more than half the sun's radiant force reaches earth, the remainder being absorbed by the atmosphere and dust which floats it; and that much of the absorption must be accomplished by the cosmic matter existing beyond the atmosphere, while that matter must be more accumulative in the neighborhood of the sun. Is it not reasonable then, to suppose that the meteoric, or cosmic dust, falling into the sun is equivalent to a dew that would cover it one-third of an inch in 24 hours; for why should not the sun attract this little amount of matter when it has power to draw worlds eighty times larger than earth and nearly 3,000 million miles distant? Astronomers do not question the power of this attraction, then if the sun can draw in one meteor may it not easily draw all that are needed to supply its heat? for the

earth's orbit around the sun is like a thread. In that orbit it passes swarms of meteors, and thus of the number that may exist in the vast circumference about our sun we have little real knowledge.

If it is true that earth receives but a portion of the fearful hydrogen heat that flames in the sun's photosphere, one-half being restrained that it does not reach the earth, may not the rest as easily be withheld in the far distant space and lie within the sun's power of gravitation?

While the sun is the light and heat of earth and all the planets and their satellites,—as a mother caring for her children,—may not all these planets, with their atmospheres and powers of gravitation, help return to her what she so freely bestows upon them? Is it more difficult to believe this than that we are daily using heat stored for ages in earth by this same wondrous sun whose light and heat, if more or less, would work the destruction of man?

When we think of the hundreds of millions of years ago that Saturn and Jupiter were a portion of the sun's body we should, according to the nebular theory, expect that those bodies would long since have cooled. But if Saturn is to-day a hot, gaseous body of not one-thousandth part the sun's mass, nor of an equal density with it, how does it happen that it is not cold? As astronomers cannot tell the years it may continue thus hot possibly there may be something about light and heat that is not yet understood? We are told by Dr. Huggins: "The green coronal line has no known representative in terrestrial substances, nor has Schuster been able to recognize any of our elements in the other lines of the corona."

It has been said that "the sun cannot shine forever;" why not? Let us imagine two persons in a room making an agreement that thereafter but one of them shall be in the room at the same time, for as one enters it the other will immediately leave. This they might agree to do every hour, day, year, or millions of years even, could they exist here, and thus keep it up forever. Likewise if the sun's rays in some manner keep returning to the sun, they may exist forever. For if true that "a particle traveling in a straight line with uniform speed in the same direction is never able to get beyond a certain limited distance from the original position, to which it will every now and then return," let us apply the theory to a ray of light and see what the result will be. One would think that a ray of light moving through a cold space

would certainly cool in one minute, but we find that such is not the case; for this ray of light is the same when it reaches earth as when it started upon its journey. The eighth minute it moved as fast and was identically the same as when first projected from the sun. Thus it ever remains flying through space at the rate of 11 millions of miles each minute, keeping the same speed as long as it can be detected by human invention, and on reaching the sun, its starting point, it must be the same in light, heat, and energy as when it left,—be that time hundreds, thousands, or millions of years,—and is ready to repeat its voyage forever; why not? If for every ray of light that goes out from the sun the same number enters, this process must forever keep the sun supplied. Is it not true of the waters of the Niagara Falls that no more flows down its stream than has already ascended to the clouds in vapor, then may it not be equally true of the sun's light and heat?

The gravitation that applies to bodies may also apply to light, and give to it as much greater an orbit than that of comets as comets have greater than that of the planets. Though light may have a vastly greater ellipse it may in the end return to the sun which projects it, to be again projected. When we think of the molecules in space that everywhere seem to possess the same properties, and the vastness of that space, we are ready to conceive that light may have the same unending properties.

Again, in reference to the stars burning out, or growing old—as is believed by some to be the case with our own sun and other suns in space, because of their varied colors and appearances—the following thoughts may be suggested. At times our sun seems to have dark spots upon its surface, while in other places great prominences are observed. This being the case who believes that everywhere the sun emits the same light and heat; and if not what must be the effect at our distance from the sun whether a dark spot, or a great projection of flaming hydrogen is directly before us? Let us imagine a sphere, with the sun for its centre, that has a radius reaching out a little way beyond earth, and a surface that might contain 1200 millions of bodies like our earth. Supposing earth to occupy, in turn, each of these 1200 millions of places we cannot believe the sun to have the same appearance from each of them. We are told by Prof. Ball that masses of vapor are frequently expelled from the interior of the sun with a speed of from 300 to nearly 1,000 miles a second, although the fact would hardly be credible only that the spectroscope enables the observer to actually witness

the ascent of these solar prominences at a distance of more than ninety millions of miles. Now from these facts would one suppose that the sun could appear the same when viewed from each of those 1,200 millions of places?

When observed from a position directly facing the dark spots the sun would seem very different from the same body viewed from a place facing the solar heights whose streams of fire were moving toward one at the rate of from 500 to 1,000 miles per second. Or let us form a sphere with the next nearest sun, Alpha Centauri, as its centre, and a radius of ten trillions of miles. In such a sphere we might place 1,210 quadrillions of earths. Who believes, that were that number of bodies of earth's size placed about the star Alpha Centauri, to each of them it would appear alike, especially if it were like our sun with dark spots and prominences upon its surface? It would seem that the light received from it might be so variable that different ages would be attributed to the sun, according to the position from which it was observed. Our own sun when seen from different points of the earth's surface—as, for instance, from the arctic region or torrid zone—does not look to us exactly the same. A very little change in the atmosphere affects the appearance of the sun as we daily view it, and the pictures of the corona taken at different places—or even at the same place with different instruments—are found on careful examination to present quite different appearances.

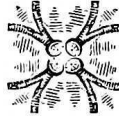
Once again, assume the sun's diameter to be one million of miles with a surface of three trillions of square miles which if two miles in depth would have twice that number of cubic miles, *i. e.*, six trillions of cubic miles. Imagine then a sphere with a radius of three billions of miles from the sun's centre, that is one reaching beyond the planet Neptune, and we have a sphere containing 108 octillions of cubic miles in volume, which divided by six trillions gives us 18,000 trillions of centuries, providing the mass has contracted six trillions of cubic miles each century. It might be said that when the sun had 6,000 millions of miles for its diameter it would have contracted more than six trillions of miles a century; but we must remember that it is a law of spherical, gaseous volumes that they revolve swifter and swifter, and grow hotter and hotter as they contract. Hence, the sun to-day being smaller is revolving more rapidly and contracting faster than ever before; though it is not detected by us. As its diameter and volume must

have been larger when it contracted more slowly its decrease could not have been more rapid, if as rapid as now, and at the rate of no more than six trillions of miles each century. If this be true we cannot have over estimated the number of centuries that the sun has been contracting. Then from these suppositions if the sun's energy has not waned in all these 18,000 trillions of centuries, it seems probable that the Power that has caused it to glow thus long may continue to give to it an energy that shall flow on with unabated strength throughout the coming ages.

As the idea of the burning-out of the sun is based upon the theory that the sun formerly was larger than now and has been reduced to its present size by contraction—although we can in no manner detect that change—the theory may still be questionable. It seems more agreeable to believe there will be no limit to the sun's bright radiance. Its unbounded flow of light throughout all the years of past time should give us assurance (until there is certain evidence to the contrary) that it is as capable of existence, and as able to resist the inroads of time, as the water in our oceans, the earth upon which we live, the air surrounding earth, and the ether above; all of which we feel exist and are preserved by the Powerful Hand and All-Seeing Eye of an Almighty Creator. It really matters little to us whether or not the sun is burning-out, for we could live, did that Creator so order, as well without as with it. There are creatures better adapted to the arctic seas than to the waters of the torrid zone; there are others that provide not for themselves but lie dormant through the cold winter months; there are birds and animals that see by night as well as by day; and our sight could as easily be adjusted for vision in one vibration per second as to make it dependent on 400 trillions of vibrations.

Still our natures are such that what would be harmful in our present state we prefer should not happen even in the years to come; and so continue to believe in the sun's endurance, although there may be some things that give credence to the idea that destruction will come to it in the future. We can understand that one ignorant of vaporization might sit at the foot of Niagara Falls and say, "Surely there cannot be water above to supply much longer this enormous, swiftly-flowing volume." In a like manner we are unable with the sun 90,000,000 miles distant, to detect any diminution of its light or heat; and judging only of the condition it was in 2,000 years ago by its power of gravitation, and its hold upon the planet-worlds—as evidenced by

the transits of Venus and the eclipses—we are led to believe that, wise as men are, they do not yet fully comprehend all the laws relating to this wonderful Solar Energy.



FOOTNOTES:

[17] Warren, “Recreations in Astronomy,” p. 94.

[18] Ball, “In Starry Realms,” p. 21.

[19] Ball, “In Starry Realms,” p. 230.

[20] Warren, “Recreations in Astronomy,” p. 82.

CHAPTER VIII.

NEBULAE.

We are told by Prof. Ball that such is the translucency of nebulae one might think to be able to see through them the stars lying in the background, were they in the right position for observation. If nebulae are thus translucent it does not seem possible that they can be composed of the same materials as the planets of the solar system, whose densities are generally so great. For were the whole of earth's substance spread out to one-hundredth of an inch in thickness we can not believe it would be sufficiently clear to allow light to penetrate it to any great extent, as earth has a density that leads one to think it is centrally composed almost—if not wholly—of iron, lead, gold, or like weighty elements that seemingly would never be transparent however highly heated and expanded. But if the nebulae are similar to the planets in their substances they must then consist of enormous masses of luminous, heated matter in a highly diffused state, in order to be perceptible to us at so great a distance; and how can they be translucent if the coloring matter is retained to darken the gases that are in combustion? Material similar to that of earth, if highly diffused and placed in an element 300° below zero, would cool almost immediately; and whether luminous or not would be likely to obstruct our view of all stars lying behind it.

We are told again by the same writer that doubtless there are hundreds, thousands, or even millions of dark bodies to each luminous one in space. But it has not been shown that these dark bodies ever needed to have been luminous in their formation, neither is it improbable that non-luminous matter,—even if once highly diffused, and spread out in the heavens as our sun is supposed once to have been diffused,—should have formed into these dark, spherical bodies in some manner similar to the condensation of vapor into clouds. If this be true the presumption is that the dark bodies are hiding many bright stars from our sight. It is not possible, then, to conceive that the many stars thus hidden—or partially hidden—may, like our sun when totally eclipsed, give a corona-like glow? Our own sun's corona flames out in every direction for more than 200 thousand miles, and should there be many suns eclipsed by the dark bodies in space might they not likewise, in

some instances, present a nebulous appearance with a startling coronal effect? Theta Orionis, the wondrous multiple star, seemingly lying in Orion's great nebula, is regarded as belonging to it because of its being in the same degree of the heavens; but although in the same degree it may be in quite a different plane, as light travels at the rate of over eleven millions of miles per minute. If this nebula is unresolvable stars then the light from it may be millions of years in reaching us, while from Theta Orionis, if it lies in the foreground, the light might reach us possibly in one hundred years,—according to the distance it is removed from earth. The probabilities are that the multiple star lies nearest the earth, for if the nebula is composed of material similar to earth, we could not think at such a distance to be able to see through a single foot, or even inch, of its substance. There are about 5,000 stars visible to the naked eye in both the northern and southern hemispheres, but only about one-third of that number are visible at any one time. Few people are able to see two thousand of them on the clearest night, while many do not see one thousand, because of their inability to detect those of the sixth magnitude. The Milky-Way, therefore, is quite like a great nebula to man's unaided vision. At each increase of photographic and telescopic power new stars are observed, until it is now said that 100 millions may be visible by their aid. If all these knowable stars are no more to space than a drop of water is to the Atlantic ocean, as the above writer has said, it is not in the least surprising that we see a great number of nebulae. This we must ever expect, even though our telescopes be increased a millionfold in power. In fact it would be most surprising if unresolvable nebulae did not forever appear in space as often as instruments of increasing penetration should be brought into positions to examine them. If space, like our oceans and our atmosphere, should be subordinate to something greater that we have never seen, and have no prospect of seeing in our present state, we can form concerning it no adequate conception.

To obtain an impression of the greatness of the nebulae in the heavens let us discard every instrument as though there were none, and upon some clear, moonless night stand gazing into the starry heavens. It might seem as though we were looking at millions of stars, but we should find by counting that the distinct points of light were only about one thousand. Imagine our sight to increase so that we beheld two thousand stars, then four thousand, eight thousand, and so on until finally, with our sight increased a thousand-fold, we were able to see one hundred millions of stars, the number that

may be seen with the most powerful instruments. What we beheld as nebulae when we saw but one thousand distinct stars would thus be resolved into shining suns with sufficient increase of power. Then is it not a fair supposition that—if our sight were adapted to the beholding—we should be able to detect not only one hundred millions of stars, but myriads of them?

Again, when we think of the irregular shapes that many of the nebulae have, can we believe that they are rotating like the sun and planets? for we look to spherical bodies for revolution. We know there are many peculiar-shaped nebulae emitting light and heat, yet that light and heat may in nowise be produced by their swift revolutions, and we cannot think of them as undergoing a change such as we would expect from the nebular theory of world-formation.

In reference to the nebula in the Sword's-handle of Orion, which contains matter sufficient to form two thousand and two hundred trillions of suns like our own; if this matter is of any density—or if not, even—it would seemingly cool almost instantly in an element 300° below zero. In truth how could it have become heated unless composed of some combustible element, like hydrogen, that would spring of itself into a mass of flame with an energy,—if there is conservation of energy,—that would exist for ever, and might repeat its work over and over again as do the oceans of our earth? There is no evidence, as far as we have ascertained, that this nebula rotates; nor can we believe that its luminosity is caused by swift revolution when our own sun, less than one million of miles in diameter, rotates but once in twenty-five days and is now moving swifter, according to the laws of mechanics appertaining to spherical, vaporous bodies, than ever before.

Even scientists and philosophers cannot tell us of the formation of a grass-seed, from which springs life identical with its kind; neither can they detect with the best microscope any difference between the varied forms of matter in the first stages of inception, nor feel sure whether there will be developed therefrom a tree, a dog, an elephant, or a man. For two hundred years *spontaneous combustion* has been discussed, some finding from an infusion of hay that life appeared; but when the idea was supposed to have been proved it was discovered that life was in *air*, and with the life-germs taken from it no spontaneous generation would arise. Later it was found that bacteriae would exist in great heat, and from that fact *biogenesis* was

thought proved; for life only could come from life. We remember also that for years the famous Bodes Law was considered fully established, until upon the discovery of planet Neptune it was found to be so far out of position that the law was no longer applicable to the distribution of the planets of the solar system. Remembering all these changes after science considered the theories well established, we believe there are things to-day about nebulae, even, that conflict with the hypothesis under consideration; as, for instance the irregular shape they often assume, and the fact that many have already been resolved into stars with the aid of sufficiently powerful instruments.



CHAPTER IX.

LIMITATIONS.

To understand how our powers are limited let us take the following illustrations. We see a horse of great fleetness, power, and intelligence, with a barn before him all his life; and yet unable to build one though he well knows it will protect him from the winds, cold, and rain. We likewise see man with grass-seed, grains of sand and oceans of water before him, and floods of air above; unable to make a particle of any of these, no matter how much he needs soil to stand on, water to quench his thirst, or air to breathe. His powers are limited; he cannot live in ocean, sail the air, or even penetrate the earth to any great depth. There are barriers against him, and although he has invented wondrous ships, they do not yet take him to the North or South Poles. Not that he does not know where the Poles are upon the earth's surface, but he seems thus far utterly unable to pass the surrounding barriers. May not these very barriers be protecting some creatures that otherwise might be exposed to injury or extermination? for we see how living things are protected, oftentimes, by their diminutive size so that animals of greater power and size cannot enter their homes to harm them; or the weaker ones have a fleetness given to them whereby they escape in time of danger.

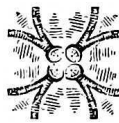
We find man endowed with such reason and wisdom that occasionally he discovers keys unlocking the mysteries of creation; these have already opened to our view, the "ancient sunlight," and developed steam, electricity, and other of the natural forces for man's benefit. Suppose that George Washington had awakened after but a night's sleep and been told that King George of England desired to speak with him. Would not the astonished General have asked: "What has become of the Atlantic ocean that I am expected to talk with King George without going to London to meet him!" That which would have inextricably puzzled the Great Commander-in-Chief is to-day scarcely thought of as mysterious, and we may believe that many like mysteries will yet yield to man's remarkable intelligence. On this great earth with its 270,000 millions of cubic miles, with its oceans of water and the ocean of air surrounding it, and with the enormous amount of light

and heat falling upon it from the sun, we see life enough to lead us to know assuredly there must be oceans of life of which we at present know nothing. Take, for instance, any field or garden and extend it however far you please—even to cover the whole earth if you will—and with but a single grass-seed you can cover the whole earth with that particular kind of grass;—assuming that it has the necessary light, heat, and moisture to make it grow and increase. We know there is light and heat in abundance, for enough falls upon the torrid zone alone to give to the whole earth a moderate temperature, if equally diffused. We know, further, that this single grass-seed is not the life but simply the key that will unlock any quantities of life. What is true of the seed is true also of every variety of vegetable and animal life. The present ocean and earth are invisible to one born blind, and as little do we see the oceans of life about us. Knowing as we must that such life exists let us look for the fire-mist of *life*, a few particles of which are familiar to us. We reason from the life we now see that it can be extended; then let us extend it, as the fire-mist of our planetary system, to Neptune's bounds, reasoning in the same manner from the least to the greatest. If we do not choose to use one cycle of time we may take millions of them, as in the cooling fire-mist theory; but we must not forget that the life we know is only on the crust of earth and goes back but a few thousand years. If earth is the same material as the sun and it takes, as we have stated, two million earths to make one sun of a million miles diameter, with no limit to the number of such suns; we can readily understand, from what we know of life here, that there may exist other great and wonderful beings beyond our highest apprehension.

We are limited in sight, for we cannot see but a few miles through the clearest atmosphere. The mountains at no great distance take on a cloud-like appearance, and resemble more nearly the surrounding sky than the great heights of rock and earth they are found to be when viewed near at hand. When we endeavor to look beneath us and find that we cannot gaze for a single foot into the earth; and that gold and diamonds might lie six inches underneath our feet and we not be able to observe them, we understand how greatly our vision is limited. Electric currents pass over the many wires strung throughout our cities, and our sight is so limited that though we look long and intensely we are not able to detect that electricity. The life in all animals and vegetation is also imperceptible to us, for we only know of its existence from the movements and appearances of bodies that possess life,

as compared with those that do not possess it. There are many qualities of life—as love, goodness, virtue, hate, jealousy, and revenge—that are as dissimilar as fire and water, or darkness and light; and could we behold them they would assume shapes differing as greatly as globes, squares, and triangles differ.

Still, what we most long to see are the spirits of our loved ones as they depart from this mortal life and ascend into the presence of their Creator; but look as long as we will, with all the faculties we possess, we must at last fall back upon the assurance that our faith in the Word of God lays hold of for our comfort and consolation. Though we there learn that “Enoch walked with God and was not, for God took him;” that Elijah went up by a whirlwind into Heaven; that Moses and Elias were revealed on the Mount of Transfiguration; and that Jesus after his resurrection was seen by His followers for forty days; yet the human beings we know and love must leave their bodies behind them as they wing their flight to worlds beyond, and our sight is so limited that we cannot perceive the souls even of our own life and being after they have left their home in the flesh. In the same manner we should know nothing of the depths of space only for the light that comes from stars removed an inconceivable distance; for of this fact we are assured by astronomers, and we should never dream that we looked a thousand miles into the heavens only for those outside worlds that give to us, through their magnitudes and distances, a faint conception of the Infinite Greatness.



CHAPTER X.

THE FIRE-MIST OF LIFE.

The theory of the formation of worlds from Nebulae is not only endorsed by Sir Robert Ball but he states his belief that the same theory is carried out upon earth in the formation of life according to the plan of Darwin. To quote his words, Darwin "has shown that the evolution of the lifeless earth from nebulae is but the prelude of an organic evolution of still greater interest and complexity." And further: "Can it be possible that the wondrous and complex phenomena known as life are purely material? Can a particle of matter which consists only of a definite number of atoms of definite chemical composition manifest any of those characters which characterize life? Take as an extreme instance the brain of an ant which is not larger than a quarter of a good-sized pin's head. It would require a volume to describe what we know of the power of ants." The following are among the wonderful things mentioned of their faculties. They communicate information to each other, build great edifices, make roads, tunnel under rivers and make temporary bridges over them by clinging together, store seeds, keep aphides as milch cows, go out to battle, and capture slaves; showing thereby a wonderful amount of power when we remember that the ant's brain is said to be but a little globule one-thousandth of an inch in diameter. From the above we learn that the ant with a brain no larger than a quarter of a pin's head is one of the most wondrous things in the world of life.

Let us conceive of such an ant standing before St. Peter's Cathedral at Rome and asking of his species: "Who made that great building?" We may anticipate the reply, "We do not know." Hear him ask again the question of all birds, insects, reptiles, and animals and receive the same answer, "We do not know." Let him ask it at last of man who replies, "I well know who made it." Encouraged he asks again, "Did you make it," but the answer comes quickly, "No." Still persistent he asks, "Could you make it?" and the answer is "No, it is not every man that could make such a building and surmount it with so wondrous a dome." Once again he asks: "But if you did not, and could not make it, how do you know the builder of it?" He receives

the answer: “I know as well as though I myself had done it. I have brain-power enough to know who planned it but it required one with a greater brain, even Michael Angelo, to conceive and build it.”

We have here seen the power of a brain smaller than a pin’s head; the additional power of one a few inches in diameter that could *know* the constructor of the great building, and yet be unable to make it; and the further increase of brain-power in the maker of that magnificent Cathedral. Yet the brain of Michael Angelo, so many times larger than the ants, is not enough for our purpose. We would find one large enough to know how the ant’s brain was formed and who formed it. Finding the brain knowing that, we would continue our queries until we found one a foot in size, a mile, earth’s size if you will, and finding that might learn what we would know of the creation of the nebulae of worlds and life; for to that brain we owe all that we possess here and may ever expect to possess.

We have been told of the wonderful instinct displayed by an ant, and yet we may not suppose, were an ant possessed of the power of speech, it could ever make observations like the above. We can conceive of a creature that possessed many times the ant’s brain-power making the inquiry, “How came this great structure?” and, asking it of a comrade that possessed the same powers as he, receiving the reply, “I do not know.” We may think of him as pushing the inquiry and receiving the answer from some that it had no author, but must have grown like the trees from nothingness; from others the conjecture that it was made by some great animal of sea or land; or again that it was made by a creature called man. We can believe that whatever their opinion might be it would matter little to Michael Angelo the author. In the vastness of the universe and the wonderful mysteries enveloping it; with the telescopes, spectroscopes, and other powerful instruments; with all the observations of centuries, and theories concerning earth’s first cause; man is yet like the ant before St. Peter’s. There is truth in the statement, “The Theory of Evolution may be true or it may be false it is still but an attempt to guess at a process; it does not touch the author of that process and never will.”^[21]

To resolve, then, the mystery of the universe we believe that a great stride would be made could we find a being with wisdom like to man’s, but possessing power to create a fire-mist such as is conceived in the Nebular Hypothesis. Finding such a being we should never for a moment think,

from what we know of dead matter, of its resolving itself into the order and system displayed in the universe, but unhesitatingly ascribe its formation to a being possessing such wisdom and power. Starting then with the theory that a Being with the intellect of man, but omnipotent power, could produce all that is now unintelligible to us; let us contemplate the appearance of His works and what we do not understand believe that He will unfold as our powers increase and as science develops.

The Revealed Word, the Great Astronomy, tells us that this Being is God and ascribes to Him the creation of the heavens, worlds, man, and life in all its forms. We can appreciate the astounding facts described in other works, and can we not *feel* the truths revealed in this? We meet people who believe this Revelation, perhaps doubtfully at first, but after study of the Word, or from hearing it explained, the truth becomes manifest to them; and what is remarkable and worthy of our contemplation is the fact that not one of the many millions who heed that Word, and live lives faithful to it, but will tell us ere they die that they do not regret the choice they have made and only wish that they had accepted its truths sooner.

The presumption is that if every human being would accept the fact that he owes his being to an Almighty God not one would ever regret it more than they who have already accepted the belief. Sooner or later all men are cast into the great fire-mist of Eternity, but ere they go hence accept or refuse a belief that *may* affect them throughout eternity. Many men are urging people to accept of the Salvation offered in the Bible, and the spirit within man feels the wisdom of such an acceptance. Men of thought see from their own anatomy that there must be a Being greater than themselves to have formed so wondrous a body, or even to have formed one of the smallest, as a grass-seed or animalcule. Why then should men need urging to make God their choice? Suppose we were suddenly cast upon a billowy sea but near us lay a life-boat, which if we laid hold upon the chances were, in nine cases out of ten, we should be saved. Would we hesitate a moment before making our choice? Desire for life would compel us to grasp the boat. Suppose, further, there was but *one* chance out of ten that if we entered the boat we should be saved, would we not instinctively make sure of that one chance, knowing if it were worthless we could be no worse off than floundering without it in the bottomless sea?

But can we look at creation, even without this Revealed Word, and say there is no evidence of a *chance* for a future life? It is few years that man lives upon earth, and those years can be but a breath to eternity; for we cannot suppose all the atoms in space will equal the years of eternity. As dying men, then, shall we live again? It is no more wonderful to believe in a new life than to believe that the combinations of our bodies are never destroyed, but are resolved again to atoms and molecules. It is no more mysterious than that the waters rise from oceans to the clouds only to return again, and repeat the process from year to year; no stranger than that earth wheels through space at the rate of nearly two millions of miles daily, without losing a drop of the waters that cover three-fourths of its surface; no more wonderful than the air that is composed of four-fifths nitrogen, and were the other fifth the same no life could exist; no more wonderful than that the sun at its great distance, holds earth as firmly by the invisible ether, as if it were an iron cord; and no more wonderful than that in earth's yearly voyage about the sun, and in the daily turning upon its axis, we do not detect the slightest jar or movement. Neither is it as mysterious as that time has no destructiveness upon earth, water, light, air, and ether, which are seemingly eternal elements; and though our spirits are imperceptible while living here, so are also some of the known energies of nature. Magnetism, that day and night, on sea or land, directs and holds the magnetic needle to the north, reveals its certain existence to us, although invisible to any faculty we possess; electricity and the gases, of which earth, air, and ocean are mainly composed, are invisible as well as indestructible; ether, filling the immensity of space, is impervious to time; and each atom of it has energy to transmit light from every sun in space, whether near or distant, without varying a second per day, thereby enabling astronomers to predict future occurrences with perfect assurance for hundreds of years.

It is no more wonderful to believe in a future life than to believe there are millions of animalcule in a drop of water; nor as mysterious as that our bodies are constantly changing their forms. The man of years has possessed many bodies that he has unconsciously cast aside, and in old age has no more the body of infancy, childhood, youth, and manhood than he has the body of another person. Yet we see man clinging to his toothless, hairless, blinded, deaf, and decrepit form while leaning upon a staff for support, as though he could not live if separated therefrom; and why not as well lay aside wholly the earthly body for the heavenly? Nearly the whole universe

is eternal, and our invisible spirits should be no more incomprehensible to us than are the universal elements and energies of nature.

It is our first living, our living now, that is wonderful and mysterious; for we find of every invention of man, the first invented one of its kind is the one most wondrous. Then with all this great universe around us shall we not live again? Are we so blinded that we see no chance of living without these bodies, when we possess faculties fitted for the contemplation of an Eternal Universe, although not the power of fully understanding its significance?

When we think of ourselves, our earth, and the sun more than a million times larger, and the size that it assumes at a distance of ninety millions of miles, while we receive but $\frac{1}{2300}$ millionth of its light and heat; when we think of the millions of suns equally great, but all, taken together, no more than a leaf in a great forest compared with the universe that embraces myriads of suns and systems; when we think again of the great nebulae, and all that may be within their range, and beyond them; we can but believe there is a God over all who may do infinitely more for us, His creatures, than all others can do, and in the eternity to come, prove our best and dearest friend.

It may be asked in this widespread universe, will He recognize earth or us? Yet we must remember there is evidence of the same formative Hand alike in each atom of air, the countless worlds, and man himself. Then who but a mightier Power can do what man can not? We may securely feel that if God made man in his own image,—which we are led to believe without the Bible's revelation—He can give to him protection, thought, and love. Then the chance, or probability—if you do not regard the possibility or certainty—of living in God's presence for as many millions of years as there are atoms in earth, which cannot comprehend eternity, are evidenced by what we here behold. Should we not live, then, in expectancy and hope, our faith grounded on what we behold in our earth, sun, and the universe?

The works of man are multiple, but among them all, we do not find one of *chance* in its formation. We always recognize man's hand and expect no form without a maker; so of the universe. The mighty worlds as well as the invisible atoms bear evidence of a Creator, and have the same assurance stamped upon them as have the inventions of man. We should thereby

recognize the truth that God exists and regulates the vast and countless worlds, the tiniest molecule, and man himself.

Surely, then, in accepting the faith of the Bible there can be no loss, while by accepting of it we may stand approved by One who created worlds, space, and all life, the existence of which is within our vision, but the contemplation of which is infinitely beyond our conception.



FOOTNOTES:

[\[21\]](#) Chambers, "Hand Book of Astronomy," preface page 9.

Transcriber's Notes

- pg 1 Changed: BIRDGEPORT, CONN.:
to: BRIDGEPORT, CONN.:
- pg 31 Changed: we can disern with a microscope
to: we can discern with a microscope
- pg 33 Changed: oceans equaling the drops of water
to: oceans equalling the drops of water
- pg 51 Changed: wasting through invisable steam
to: wasting through invisible steam
- pg 63 Changed: flow of light througout
to: flow of light throughout
- pg 65 Changed: spread out to one-hundreth of an inch
to: spread out to one-hundredth of an inch
- pg 76 Changed: things mentioned of their faculities
to: things mentioned of their faculties
- pg 81 Changed: electrcity and the gases
to: electricity and the gases

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